

# Examen tipo 2<sup>do</sup> Parcial:

1) Para el S.E.C.:

Determinar:

$$\begin{cases} x_1 - x_2 + 2x_3 + 3x_4 = -2 \\ 2x_1 - x_2 + 3x_3 + 5x_4 = -2 \\ -3x_1 + 4x_2 - 7x_3 - 10x_4 = 8 \end{cases}$$

i) la solución general

ii) la solución general del S.E.C. Homogeneo asociado

Llevemoslo a la matriz aumentada:

$$\left[ \begin{array}{cccc|c} 1 & -1 & 2 & 3 & -2 \\ 2 & -1 & 3 & 5 & -2 \\ -3 & 4 & -7 & -10 & 8 \end{array} \right] \xrightarrow{\substack{-2F_1 + F_2 \\ +3F_1 + F_3}} \left[ \begin{array}{cccc|c} 1 & -1 & 2 & 3 & -2 \\ 0 & 1 & -1 & -1 & 2 \\ 0 & 1 & -1 & -1 & 2 \end{array} \right] \xrightarrow{\substack{-1F_2 + F_3 \\ F_2 + F_1}} \left[ \begin{array}{cccc|c} 1 & 0 & 1 & 2 & 0 \\ 0 & 1 & -1 & -1 & 2 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right]$$

Volvemos a llevar a términos de variables

Despejados:

$$\begin{aligned} x_1 + \cancel{0x_2} + x_3 + 2x_4 &= 0 \\ \cancel{0x_1} + x_2 - x_3 - x_4 &= 2 \end{aligned} \quad \begin{aligned} x_1 &= -x_3 - 2x_4 \\ x_2 &= 2 + x_3 + x_4 \end{aligned} \quad ; \quad x_3, x_4 \in \mathbb{R}$$

$$x_3 = t$$

$$x_4 = s$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} -t - 2s \\ 2 + t + s \\ t \\ s \end{bmatrix} = \begin{bmatrix} 0 \\ 2 \\ 0 \\ 0 \end{bmatrix} + t \begin{bmatrix} -1 \\ 1 \\ 1 \\ 0 \end{bmatrix} + s \begin{bmatrix} -2 \\ 1 \\ 0 \\ 1 \end{bmatrix}$$

↓  
Solución particular

↓  
Sol. Sistema homogéneo

2) Dadas las matrices:

$$A = \begin{bmatrix} 1 & 2 & -3 \\ 1 & 1 & -2 \\ 0 & 1 & -2 \end{bmatrix}_{3 \times 3}, \quad B = \begin{bmatrix} 1 & 1 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}_{3 \times 3}, \quad C = \begin{bmatrix} 1 & -2 \\ 0 & 3 \\ 3 & 1 \end{bmatrix}_{3 \times 2}$$

Calcule:

$$\begin{aligned} i) A^{-1} &= \left[ \begin{array}{ccc|ccc} 1 & 2 & -3 & 1 & 0 & 0 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 0 & 1 & -2 & 0 & 0 & 1 \end{array} \right] \xrightarrow{-1F_1 + F_2} \left[ \begin{array}{ccc|ccc} 1 & 2 & -3 & 1 & 0 & 0 \\ 0 & -1 & 1 & -1 & 1 & 0 \\ 0 & 1 & -2 & 0 & 0 & 1 \end{array} \right] \xrightarrow{\substack{-1F_2 \\ -2F_2 + F_1 \\ -1F_2 + F_3}} \left[ \begin{array}{ccc|ccc} 1 & 0 & -1 & -1 & 2 & 0 \\ 0 & 1 & -1 & 1 & -1 & 0 \\ 0 & 0 & -1 & 1 & 1 & 1 \end{array} \right] \\ &\quad \xrightarrow{\substack{-1F_3 \\ 1F_3 + F_2}} \left[ \begin{array}{ccc|ccc} 1 & 0 & 0 & 0 & 1 & -1 \\ 0 & 1 & 0 & 2 & -2 & -1 \\ 0 & 0 & 1 & 1 & -1 & -1 \end{array} \right] \end{aligned}$$

$$ii) A + 3B - CC^T$$

$$\begin{bmatrix} 1 & 2 & -3 \\ 1 & 1 & -2 \\ 0 & 1 & -2 \end{bmatrix} + 3 \begin{bmatrix} 1 & 1 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix} - \begin{bmatrix} 1 & -2 \\ 0 & 3 \\ 3 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 & 3 \\ -2 & 3 & 1 \end{bmatrix}_{3 \times 2}$$

$$\begin{bmatrix} 1 & 2 & -3 \\ 1 & 1 & -2 \\ 0 & 1 & -2 \end{bmatrix} + \begin{bmatrix} 3 & 3 & -3 \\ 0 & 3 & -3 \\ 0 & 0 & 3 \end{bmatrix} - \begin{bmatrix} 5 & -6 & 1 \\ -6 & 9 & 3 \\ 1 & 3 & 10 \end{bmatrix}$$

$$\begin{bmatrix} -1 & 11 & -7 \\ 7 & -5 & -8 \\ 1 & -2 & -9 \end{bmatrix}$$

$$iii) (C^T C)^{-1}$$

$$\begin{bmatrix} 1 & 0 & 3 \\ -2 & 3 & 1 \end{bmatrix}_{3 \times 2} \begin{bmatrix} 1 & -2 \\ 0 & 3 \\ 3 & 1 \end{bmatrix}_{2 \times 3} = \begin{bmatrix} 10 & 1 \\ 1 & 14 \end{bmatrix}$$

$$\text{Inversa de } C^T \cdot C = \frac{1}{\det(C^T \cdot C)} \cdot (\text{adj } C^T \cdot C)$$

$$(C^T \cdot C)^{-1} = \frac{1}{139} \begin{bmatrix} 14 & -1 \\ -1 & 10 \end{bmatrix} = \begin{bmatrix} 14/139 & -1/139 \\ -1/139 & 10/139 \end{bmatrix}$$

3) a) Sea  $A = \begin{bmatrix} a & b & c \\ x & y & z \\ \alpha & \beta & \delta \end{bmatrix}$  Sabiendo que  $\det(A) = -3$  Calcule:

$$i) \det \begin{bmatrix} 2x & 2y & 2z \\ -a & -b & -c \\ 3\alpha & 3\beta & 3\delta \end{bmatrix} \xrightarrow{F_2 \leftrightarrow F_1} \begin{bmatrix} 2x & 2y & 2z \\ -a & -b & -c \\ 3\alpha & 3\beta & 3\delta \end{bmatrix} = 2 \cdot 3 \det \begin{bmatrix} a & b & c \\ x & y & z \\ \alpha & \beta & \delta \end{bmatrix} = 6 \cdot -3 = -18$$

$$ii) \det \begin{bmatrix} a & 2x & 2 \\ b & 2y & \beta \\ c & 2z & \delta \end{bmatrix} = \det \begin{bmatrix} a & b & c \\ 2x & 2y & 2z \\ \alpha & \beta & \delta \end{bmatrix} = 2 \cdot \det \begin{bmatrix} a & b & c \\ x & y & z \\ \alpha & \beta & \delta \end{bmatrix} = 2 \cdot -3 = -6$$

$$iii) \det(5A^{-1}) = 5^3 \cdot \det A^{-1} = 5^3 \cdot \frac{1}{-3} = -\frac{125}{3}$$

$$iv) \det \begin{bmatrix} a & b & c \\ x+2a & y+2b & z+2c \\ 2\alpha & 2\beta & 2\delta \end{bmatrix} = \det \begin{bmatrix} a & b & c \\ x & y & z \\ 2\alpha & 2\beta & 2\delta \end{bmatrix} = 2 \det \begin{bmatrix} a & b & c \\ x & y & z \\ \alpha & \beta & \delta \end{bmatrix} = 2 \cdot -3 = -6$$

b) Muestre que  $\det \begin{bmatrix} 1-\lambda & 1 & 1 \\ 1 & 1-\lambda & 1 \\ 1 & 1 & 1-\lambda \end{bmatrix} = \lambda^2(3-\lambda)$

$$\det \begin{bmatrix} 3-\lambda & 1 & 1 \\ 3-\lambda & 1-\lambda & 1 \\ 3-\lambda & 1 & 1-\lambda \end{bmatrix} = (3-\lambda) \det \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1-\lambda & 1 \\ 1 & 1 & 1-\lambda \end{bmatrix}$$